CalorieTrack

Track your calories smarter with machine learning.

Problem Statement

Tracking daily calorie intake is essential for maintaining a healthy lifestyle, but it can be time-consuming and inaccurate when done manually. Many individuals struggle to estimate their caloric needs based on age, height, weight, and activity levels. This gap highlights the need for an intelligent solution that simplifies calorie tracking while providing accurate predictions tailored to individual profiles.

Features

Personalized Calorie Predictions: Uses AI to predict daily calorie needs based on age, height, weight, and activity duration.

Data-Driven Insights: Visualizes trends in calorie consumption and offers actionable insights.

Efficient Machine Learning Models: Integrates advanced models like XGBoost and Random Forest for accurate predictions.

User-Friendly Interface: Simplifies input of personal details and activity duration for effortless tracking.

Real-Time Feedback: Provides immediate predictions and comparisons to actual calorie requirements.

Adaptable Across Profiles: Supports multiple user demographics with gender encoding and flexible data inputs.

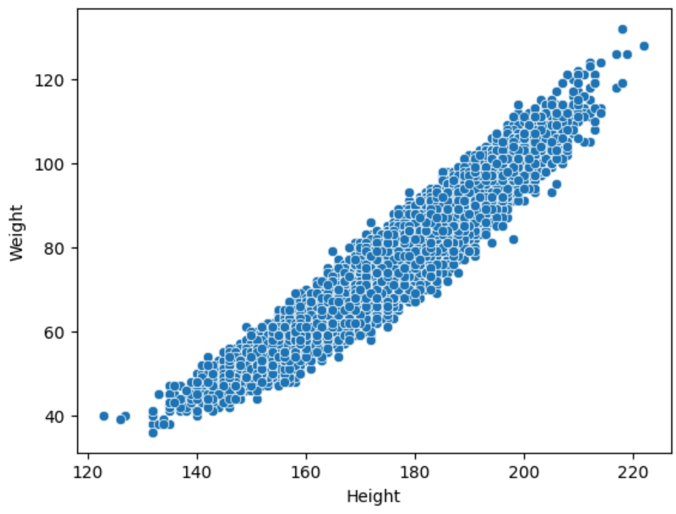
Scalable for Future Features: Ready for integration with wearables and dietary recommendation systems.

Data Workflow

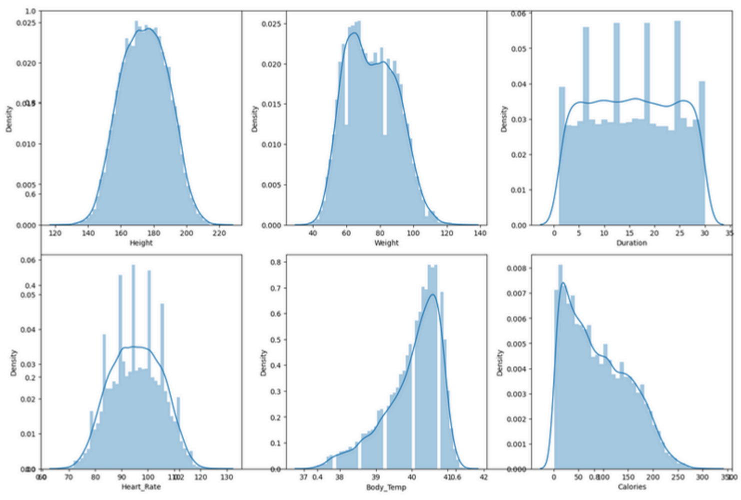
Data Collection → Data Cleaning → Feature Engineering → Model Training → Predictions & Insights

1. **Data Collection:** Data collected from a combined dataset of user demographics, activity levels, and calories.
2. **Data Cleaning:** Handled missing values and encoded gender for analysis readiness.
3. **Feature Engineering:** Generated meaningful features like Age, Height, Weight, and Activity Duration.
4. **Model Training:** Trained advanced ML models like XGBoost and Random Forest for accurate predictions.
5. **Predictions and Insights:** Generated calorie predictions and visualized trends for personalized insights.

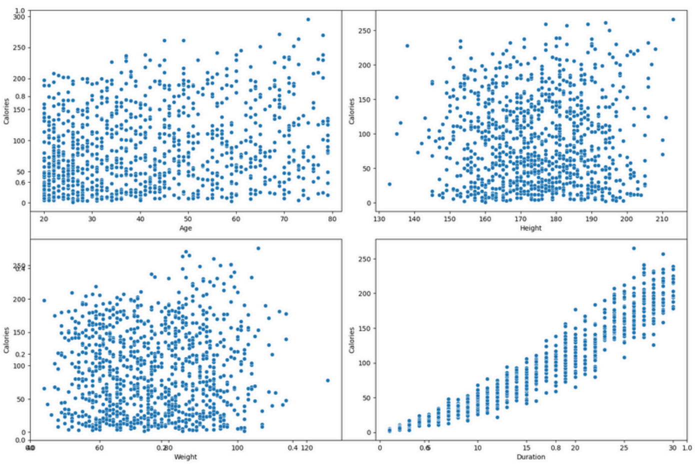
Results



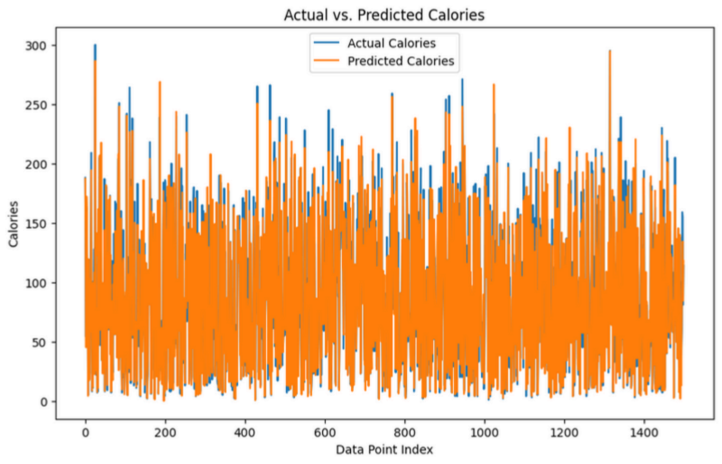
The scatterplot visualizes the relationship between height and weight, indicating a positive correlation.



Histograms visualize the distributions of numerical features, such as age, height, weight, duration, heart rate, and body temperature.



Scatterplots explore the relationships between age, height, weight, duration, and calories burned during exercise.



The line graph compares actual calorie values with those predicted by the best performing model, demonstrating its accuracy.

Tools & Technologies

scikit-learn

pandas

Python

XGBoost

seaborn

Call to Action

Discover smarter calorie tracking with our web based AI-powered app! Our app leverages machine learning to help you track your calories more accurately and make better dietary choices. Learn more at by scanning the QR code to explore the full capabilities of the app and see how it can improve your health. Try our demo today! Get a firsthand experience of the features by accessing the demo version now.

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